See the Coarse, Learn the Fine: Resolution-Adaptive Hybrid Segmentation and Regression for Soil Grain Size Distributions from Smartphone Photos

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Abstract

For the Kaggle challenge on soil grain size from photos [5], we propose to measure *the grains a phone camera can resolve with a frozen Segment Anything Model [6] and* learn *the fraction it cannot, conditioned on the measured grains. The two meet at a resolution-adaptive crossover sieve and are blended by the measured coarse evidence into a curve that is monotonic by construction.*

1. Problem Definition

Task. Given one or more top-down photos of a soil sample spread in a tray, predict the cumulative mass percentage passing at the eleven DIN EN ISO 14688-1 sieve sizes (mm) $x_i \in \{0.002, 0.0063, 0.02, 0.063, 0.2, 0.63, 2, 6.3, 20, 63, 200\}$. The output must lie in $[0, 100]$, never decrease, and equal 100 at 200 mm. Submissions are scored with the log-weighted Earth Mover's Distance (EMD), the area between the true and predicted curves on a log axis:

$$\mathrm{EMD}(P, \hat{P}) = \sum_{i=1}^{10} \left|P_i - \hat{P}_i\right| \left(\log_{10} x_{i+1} - \log_{10} x_i\right). \quad (1)$$

Data and challenges. Training has only **24 soil samples** (127 photos, Motorola and Samsung); the test set has 10 samples (35 photos, iPhone 14/16). Three properties make it hard. (1) *Resolution floor:* rescaling the given pixels-per-mm to the released image sizes puts training photos at $\rho \approx 4.6$ px/mm, so a 2 mm grain spans 9 px and every sieve ≤ 0.63 mm is under 3 px (Tab. 1). Yet the median training sample has 63% of its mass finer than 2 mm, and seven of the ten equally weighted EMD terms sit at sieves the camera cannot resolve. (2) *Domain shift:* test photos come from unseen phones and are sampled 3–4$\times$ more finely. (3) *Tiny data:* end-to-end deep regression on 24 soils will overfit. The organisers' constant baseline scores an EMD of 102.0 on the training labels and the mean training curve 89.9 in leave-one-soil-out cross-validation (our computation).

Sieve (mm)	0.063	0.2	0.63	2	6.3	20
Train, 4.6 px/mm	0.3	0.9	2.9	9.2	**29**	92
Test iPhone 14, 13.9	0.9	2.8	8.8	**28**	88	278
Test iPhone 16, 19.5	1.2	3.9	12.3	**39**	123	390

Table 1. Grain width in pixels at each sieve size. SAM-style segmentation is reliable from about 25–30 px [3]. Bold marks the adaptive crossover d_c.

2. Literature Review

Direct regression maps the whole image straight to the curve: SediNet [2] and GRAINet [7] regress percentiles of sediment and gravel distributions with CNNs. PSDNet [9] trains on synthetic packings; Coenen *et al*. [4] skip segmentation because small particles cannot be identified. Closest to this competition, GRAI1 [20] and GRAI3 [19] regress the two parameters of a Weibull curve from close-range and smartphone soil photos. Texture methods such as autocorrelation [17] and wavelets [1] estimate grain size without delineating any grain. Here every point on the curve is inferred, even for clearly visible coarse grains.

Segmentation. SAM [6, 15] has made zero-shot grain delineation practical, as in OrthoSAM [3], ImageGrains 2.0 [8] and Segmenteverygrain [21]. Xiao *et al*. [22] map SAM mask statistics to coarse-aggregate size distributions with an ANN, but they do not model fines. Youwai and Makam [23] show segmentation-only analysis fails below 2 mm. Rock-fragmentation analysis instead fits a Rosin–Rammler [16] or Swebrec [12] curve to the measured coarse fragments and extrapolates it into the fines, assuming they share its shape.

Gap. To our knowledge, no prior work combines foundation-model measurement of the resolvable fraction with a *learned* model of the unresolved fraction that is conditioned on the measured grains, under a crossover that adapts to image resolution.

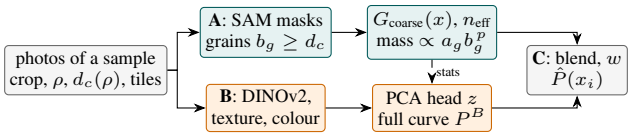


Figure 1. Pipeline. Path A measures grains at or above the crossover d_c. Path B learns the whole curve, using statistics from Path A as extra inputs. Path C blends the two coarse shapes above d_c.

3. Proposed Approach

Preprocessing and adaptive crossover. We crop each photo to the tray and compute its true scale ρ (full-resolution px/mm times released/full width). The crossover is the smallest sieve that SAM can reliably resolve at that scale:

$$d_c(\rho) = \min\{x_i : \rho\, x_i \geq \tau\}, \qquad \tau \approx 25 \text{ px}. \tag{2}$$

This gives d_c=6.3 mm for the training photos and 2 mm for the test photos (Tab. 1). Large photos are tiled at 1024 px with overlap.

Path A: measured coarse fraction (training-free). SAM/SAM 2 automatic masks are generated per tile and merged across overlaps. We drop the tray background, masks below τ, low-solidity masks (clumps, touching grains), and clods whose colour matches the fine matrix. A grain lying flat shows its long axis a_g and intermediate axis b_g, and sieves sort by b_g, so we take b_g as the minimum Feret diameter in mm. With thickness $\propto b_g$, mass is $m_g \propto a_g b_g^p$ with p=2 by default. Masks from *all photos of a sample* are pooled, since one photo shows few large grains:

$$G_{\text{coarse}}(x) = \frac{\sum_{g:\, b_g \leq x} m_g}{\sum_g m_g}, \qquad x \geq d_c. \tag{3}$$

Path B: learned curve. Path B predicts the whole curve, so it does not depend on where the crossover falls. The curve is a composition π over the eleven sieve-bounded size classes; to suit 24 soils, the head predicts only r=2–3 coefficients z in a PCA basis (μ, V) of the log-ratio GRAI1 curves:

$$\pi = \text{softmax}(\mu + Vz), \qquad P^B(x_j) = 100 \textstyle\sum_{k \leq j} \pi_k. \tag{4}$$

Every z gives a monotone curve and z=0 gives the mean training curve, so a regularised head falls back to the mean-curve baseline rather than overfitting. We write $\hat{F} = P^B(d_c)$. Inputs are (i) global image features, (ii) texture statistics (autocorrelation and wavelet [1, 17]), (iii) colour statistics in CIELAB space, and (iv) *coarse statistics from Path A*: resolved grain count, the fraction of the tray covered by masks, and the coarse D_{50}. Input (iv) separates us from parametric extrapolation: resolved grains covering only 30% of the tray imply the rest is finer (a learned cue, since area is not mass). For train/test consistency, (iv) uses only grains ≥6.3 mm (visible at every resolution), and Path B features are extracted after resampling every image to 4.6 px/mm. For (i) we use frozen DINOv2 [11] tile embeddings averaged over the sample and PCA-reduced. The head is a ridge regressing the PCA scores of the true curve, or a small MLP minimising Eq. (1) on the *stitched* curve.

Path C: evidence-weighted stitching. Let $G_B(x) = (P^B(x) - \hat{F})/(100 - \hat{F})$ be Path B's own coarse shape. Below d_c, $\hat{P} = P^B$; above it the two shapes are blended:

$$\hat{P}(x) = \hat{F} + (100 - \hat{F})\big[w\, G_{\text{coarse}}(x) + (1-w)\, G_B(x)\big], \tag{5}$$

with $w = n_{\text{eff}}/(n_{\text{eff}} + \kappa)$ and $n_{\text{eff}} = (\sum_g m_g)^2 / \sum_g m_g^2$ the effective grain number, small when one pebble dominates. With no resolved grains w=0 and the curve is pure Path B; w=1 is a hard stitch. The result is continuous, monotonic and reaches 100 at 200 mm by construction.

Datasets. The Kaggle training soils [5] come from the public GRAI1 release [18], which adds two soils (one clay-rich) and the raw PSD curves, with no overlap with the test samples. The rules allow free public external data, so we add (i) FRDR Photogranulometry [13, 14] (12,714 photos of 321 Montreal soils, fixed lab rig, ∼25 px/mm, ASTM sieves) and (ii) SandSnap [10], phone photos of beach sand with a coin for scale and sieved d_{10}–d_{90}. We (a) resample all photos to Kaggle scales, (b) map labels onto the DIN sieves by monotone PCHIP interpolation in $\log x$, masking sieves with no label, and (c) pretrain Path B on (i)–(ii).

Evaluation. Leave-one-soil-out cross-validation on the 24 Kaggle soils (nested to tune r and the ridge penalty) plus the leaderboard; we report mean EMD with paired bootstrap intervals, per-sieve error and $D_{10}/D_{50}/D_{60}$ errors, split into gravelly and fine-grained soils. *Resolution simulation:* FRDR photos downsampled to 4.6, 13.9 and 19.5 px/mm validate Path A at d_c=2 mm, set τ, p and κ, and compare adaptive vs. fixed d_c, which the Kaggle training photos cannot test. *Baselines:* constant and mean-curve predictors; GRAI-style Weibull regression on the same backbone; SAM only; SAM with Rosin–Rammler/Swebrec extrapolation; texture-only regression. *Ablations:* $w \in \{0, 1, \text{adaptive}\}$ (pure regression, hard stitch, ours); PCA head vs. free softmax increments; Path B with vs. without coarse statistics; with vs. without external pretraining; count, area or mass weighting. *Qualitative results:* mask overlays and predicted vs. true curves. Kaggle submissions and baselines come first (closes ∼Nov. 30), then pretraining and ablations.

4. Team Backgrounds and Outside Work

Haruki Matsuo has done LightGBM feature engineering and regression (Kaggle house prices) and hand-landmark shape/motion features for Japanese Sign Language recognition (hackathon), both relevant to extracting grain features and regressing the curve. **Erhans Arya Kencana** and **Jason Supala** have no related outside work. The project is not part of another course.

References

[1] Daniel Buscombe. Transferable wavelet method for grain-size distribution from images of sediment surfaces and thin sections, and other natural granular patterns. *Sedimentology*, 60(7):1709–1732, 2013. 1, 2

[2] Daniel Buscombe. SediNet: A configurable deep learning model for mixed qualitative and quantitative optical granulometry. *Earth Surface Processes and Landforms*, 45(3): 638–651, 2020. 1

[3] V. Chan, A. Rheinwalt, and B. Bookhagen. OrthoSAM: Multi-scale extension of the Segment Anything Model for river pebble delineation from large orthophotos. *Earth Surface Dynamics*, 14(3):391–416, 2026. 1

[4] Max Coenen, Dries Beyer, Christian Heipke, and Michael Haist. Learning to sieve: Prediction of grading curves from images of concrete aggregate. *arXiv preprint arXiv:2204.03333*, 2022. 1

[5] Kaggle. Predicting soil grain size distributions from images. Kaggle competition, https://www.kaggle.com/competitions/soil-grain-size-from-photos, 2026. 1, 2

[6] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *ICCV*, 2023. Code: https://github.com/facebookresearch/segment-anything. 1

[7] Nico Lang, Andrea Irniger, Agnieszka Rozniak, Roni Hunziker, Jan Dirk Wegner, and Konrad Schindler. GRAINet: Mapping grain size distributions in river beds from UAV images with convolutional neural networks. *Hydrology and Earth System Sciences*, 25:2567–2597, 2021. 1

[8] D. Mair, G. Witz, A. Do Prado, et al. ImageGrains 2.0: Improved precision and generalization for grain segmentation. *Earth Surface Dynamics*, 14:527–551, 2026. 1

[9] Javad Manashti, Pouyan Pirnia, Alireza Manashty, Sahar Ujan, Matthew Toews, and François Duhaime. PSDNet: Determination of particle size distributions using synthetic soil images and convolutional neural networks. *arXiv preprint arXiv:2303.04269*, 2023. 1

[10] Brian McFall, Daniel Buscombe, Shelley Whitmeyer, David Young, and Shannon Stever. The SandSnap project: 2020–2021 sieved grain-size data and associated sediment imagery. Zenodo, https://doi.org/10.5281/zenodo.7063226, 2022. 2

[11] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. Code: https://github.com/facebookresearch/dinov2. 2

[12] Finn Ouchterlony. The Swebrec function: Linking fragmentation by blasting and crushing. *Mining Technology*, 114(1): 29–44, 2005. 1

[13] Thomas Plante St-Cyr, François Duhaime, Jean-Sébastien Dubé, and Simon Grenier. Photogranulometry – dataset of soil images with corresponding particle size distributions. Federated Research Data Repository, https://doi.org/10.20383/103.01316, 2025. 2

[14] Thomas Plante St-Cyr, François Duhaime, Jean-Sébastien Dubé, and Simon Grenier. Dataset of soil images with corresponding particle size distributions for photogranulometry. *arXiv preprint arXiv:2506.17469*, 2025. 2

[15] Nikhila Ravi et al. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. Code: https://github.com/facebookresearch/sam2. 1

[16] P. Rosin and E. Rammler. The laws governing the fineness of powdered coal. *Journal of the Institute of Fuel*, 7:29–36, 1933. 1

[17] David M. Rubin. A simple autocorrelation algorithm for determining grain size from digital images of sediment. *Journal of Sedimentary Research*, 74(1):160–165, 2004. 1, 2

[18] Enrico Soranzo. Dataset of close-range soil images and corresponding particle size distributions. *Data in Brief*, 60:111631, 2025. Data: https://doi.org/10.5281/zenodo.14725633. 2

[19] Enrico Soranzo. GRAI3: Generalizable soil particle size distribution from smartphone images. *Computers and Geotechnics*, 199:108362, 2026. 1

[20] Enrico Soranzo, Carlotta Guardiani, and Wei Wu. Convolutional neural network prediction of the particle size distribution of soil from close-range images. *Soils and Foundations*, 65(1):101575, 2025. 1

[21] Zoltán Sylvester. Segmenteverygrain. GitHub, https://github.com/zsylvester/segmenteverygrain. 1

[22] Y. Xiao, Y. Peng, M. Wang, Y. Ning, Y. Zhou, K. Kong, and Y. Long. A novel method for predicting coarse aggregate particle size distribution based on segment anything model and machine learning. *Construction and Building Materials*, 429:136429, 2024. 1

[23] Sompote Youwai and Parchya Makam. Computer vision for particle size analysis of coarse-grained soils. *arXiv preprint arXiv:2311.06613*, 2023. 1